

BISA SAITEJASWINI RAO

Robotics & Automation Engineer

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My Portfolio- https://saitejaswinirao.github.io/

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Career Objective

Robotics and Automation Engineer with hands-on experience in ROS, autonomous robots, RPA, PLC, and embedded systems. Skilled in robot design, control systems, computer vision, and automation, OpenCV, Arduino, and UiPath. Seeking to learn and enhance system efficiency, product quality, and operational performance while supporting organizational innovation and growth.

Education

Aug 2025 – Present
Gandhinagar, Gujarat
International Professional Master's Program in Advanced Robotics Technology
New Age Makers' Institute of Technology

Feb 2023 – Jun 2025
Guntur, Andhra Pradesh
B.Tech (Honours) in Product Design **8.51**
Vignan's Foundation for Science, Technology & Research

Aug 2022 – Jun 2025
Guntur, Andhra Pradesh
Bachelor in Technology in Robotics & Automation **8.45**
Vignan's Foundation for Science, Technology & Research

May 2019 – Jun 2022
Hyderabad, Telangana
Diploma in Automation & Robotics Engineering **7.78**
Central Institute of Tool Design (MSME)

Mar 2018 – Apr 2019
Ghatkesar, Telangana
Central Board of Secondary Education (Xth) **7.75**
Vignan Prabodhananda Prashanti Niketan

Experience

Dec 2024 – May 2025
Guntur, Andhra Pradesh
Teaching Assistantship
Vignan's University (VFSTR)

- Delivered and supported coursework in **Fundamentals of Robotics, Mobile Robotics, Robotics for Engineers, and Engineering Drawing**, integrating theory with practical learning.
- Conducted **hands-on labs and tutorials** on **robot kinematics & mobile robot concepts**.
- Mentored students in **robot design, simulation, and project development** using **Fusion 360**, including debugging and performance evaluation.

Oct 2023 – Nov 2023
Hyderabad, Telangana
RPA Practitioner Intern
Henry Harvin Education

- Automated **10+ routine workflows** using UiPath, boosting process efficiency.
- Identified **multiple automation opportunities** in collaboration with production teams.
- Tested, deployed & monitored **high-reliability bots**, ensuring smooth performance.

Projects

Oct 2025 – Present
Development of Teleoperation Controller Module

- A 3D-printed enclosure for housing and protection.
- Three control modes – Unilateral, Bilateral, and Virtual Wall.

Oct 2025 – Present
Fabrication & Control of Soft Gripper

- Pneumatically actuated
- Closed-loop control using a pressure sensor

Sep 2025 – Present
Design & Control of Warehouse Mobile Robot

- LIDAR-based navigation
- Designed a material dispensing unit with a planar manipulator

Feb 2024 – May 2025
Autonomous Rover with Rocker-Bogie Mechanism

- ROS-based control
- Fabricated using FDM technology (PLA & ABS)

Nov 2023 – Jun 2025 **A Semi-Humanoid Robot with Autonomous Capabilities**
- Facial recognition through OpenCV
- Arms with 6-DOF

Skills

Programming Languages: C, Python, HTML

Development Boards: Arduino Uno/Mega, Raspberry Pi, Jetson Nano

Technical: ROS, RPA, CAD Modeling, PLC, HMI, Pneumatics, OpenCV, Robot System Integraion (Fanuc)

Soft Skills: Communication, Project Management, Adaptability, Initiative, Leadership, Teamwork

Courses

Jul 2024 – Oct 2024 **Metal Additive Manufacturing**
NPTEL

Aug 2023 – Oct 2023 **RPA Using UiPath Course**
NASSCOM

Jan 2023 – Apr 2023 **The Joy of Computing Using Python**
NPTEL

Certificates

Certified Robotic Process Automation Practitioner, Henry Harvin -2023

Preliminary English Test (PET) - B1 Level, Cambridge University -2023

Business English Certificate Vantage (BEC) – B2 Level, Cambridge University -2024

SOLIDWORKS Design Associate (CSWA), Dassault Systèmes -2026

Publications

11/2024 **The Impact of Robots in Agriculture for Enhanced Precision in Farming**
IJRES Journal

Patents

02/2026 Autonomous Space Exploring Rover with onboard Manipulator
(Under Review - Ref. No. 490026-001)

02/2026 Autonomous Localization and Mapping Robot for Delivery Applications
(Under Review - Ref. No. 489955-001)

02/2026 Autonomous House Cleaning Robot
(Under Review - Ref. No. 489952-001)

02/2026 Autonomous Precision Weed Detection and Removal Robot for Smart Farming
(Under Review - Ref. No. 490033-001)

Achievements

Secured "AIR 2" in Janatics Skill Automation Competition

Cleared “Quals Round” in IRoCU – 2024

Received “Student SEED Grant” of Rs. 1.9 Lakhs by Vignan’s University

Accomplished 'C' certificate in “National Cadet Corps (NCC)”

Languages

English | Hindi | Telugu | Marathi